

CHRISTOPHER J. SNELLING

Columbia, MD | christopher.j.snelling@gmail.com

chrisjs23.github.io/personal-portfolio | linkedin.com/in/christopher-snelling

PROFESSIONAL SUMMARY

Software engineer and M.S. Computer Science candidate with hands-on experience building machine-learning and natural-language-processing systems, data and evaluation pipelines, games, compilers, and self-hosted infrastructure. Proficient in Python, Java, C/C++/C#, SQL, and modern development tools, with a strong foundation in object-oriented design, algorithms, debugging, testing, and technical documentation. Experienced in translating complex requirements into practical, usable systems and collaborating across technical and instructional teams.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, C#, SQL, Bash, GDScript, Lua

Machine Learning & Data: NumPy, pandas, matplotlib, scikit-learn, PyTorch, Hugging Face Transformers, natural language processing, data preprocessing, model evaluation, cross-validation

Frameworks & Platforms: libGDX, Godot, Unity, MongoDB, Jellyfin, Raspberry Pi, Flex, Bison, MakeMKV

Development Tools: Linux, Git/GitHub, Jupyter, VS Code, command-line tooling, REST APIs

Software Engineering: Object-oriented programming, algorithms and data structures, debugging, unit testing, Agile/Scrum collaboration, accessibility/WCAG, technical writing and documentation

EDUCATION

M.S. Computer Science, Artificial Intelligence Focus | Johns Hopkins University **Expected December 2026**

B.S. Computer Science | University of Maryland Global Campus **2023**

GPA: 3.7

B.S. Psychology | Virginia Polytechnic Institute and State University **2013**

PROFESSIONAL EXPERIENCE

Course Support Assistant | Johns Hopkins University, Center for Teaching and Learning Design **2025-Present**

- Support faculty, instructional designers, and course support specialists in creating WCAG-compliant learning materials for online courses across multiple disciplines.
- Convert outdated course materials into modern, accessible formats while preserving instructional content and usability.

Delivery Driver | DoorDash **2025**

- Independently managed real-time logistics, route changes, and customer communication to deliver orders accurately and on time while maintaining strong customer ratings.

Full-Time Family Caregiver **2013-2023**

- Provided full-time care for an ill family member, requiring resilience, self-direction, disciplined time management, and coordination of changing daily needs.

Student Research Assistant | Department of Computer Science, Virginia Tech **2012**

- Coordinated multi-session user-experience studies, prepared hardware and software laboratories, and processed experimental datasets for analysis.

TECHNICAL PROJECTS

Complex Systems: Kuramoto Associative Memory **July 2026**

Python | NumPy | matplotlib

- Implemented a 64-oscillator associative-memory network in which phase configurations encoded 8x8 binary patterns and Hebbian coupling stored target memories.
- Ran repeated simulations across coupling strengths and corruption levels, measuring pixel accuracy, pattern overlap, exact retrieval rate, and retrieval behavior over time.
- Visualized successful, partial, and failed retrieval cases to show how collective synchronization reconstructs corrupted patterns and where recovery breaks down.

Deep Learning: BERT Robustness Under Domain Shift

April 2026

Python | PyTorch | Hugging Face Transformers

- Fine-tuned a BERT sentiment classifier on balanced IMDb reviews and evaluated generalization to Amazon product reviews with different vocabulary, structure, and domain conventions.
- Compared full fine-tuning with layer freezing, increased regularization, and synonym-based data augmentation using accuracy, F1 score, ROC-AUC, and expected calibration error.
- Achieved 92.19% accuracy on IMDb and 91.69% on Amazon; found that baseline BERT transferred well while heavier layer freezing improved calibration at a modest cost to predictive performance.

Natural Language Processing: ADHD Discourse Detection

December 2025

Python | pandas | scikit-learn | sentence-transformers

- Built binary text-classification systems distinguishing ADHD and control Reddit posts using TF-IDF with Linear SVM and MiniLM sentence embeddings with Linear SVM.
- Designed stratified train, validation, and test splits with macro-F1 evaluation to compare lexical and contextual representations consistently.
- Conducted keyword-masking robustness analysis showing that contextual embeddings degraded less than lexical models when explicit ADHD terminology was removed.

Reinforcement Learning: Racetrack Control

December 2025

Python | NumPy | pandas

- Implemented Value Iteration, Q-Learning, and SARSA in a stochastic racetrack environment with a velocity-based state space, collision detection, and alternative crash-handling rules.
- Designed experiments across track geometries and crash penalties to compare convergence, reliability, and on-policy versus off-policy behavior.
- Demonstrated safety-performance tradeoffs, with Q-Learning favoring higher-risk routes and SARSA learning more conservative policies under harsh penalties.

Neural Networks and Autoencoder Pretraining

November 2025

Python | NumPy | pandas

- Implemented linear and logistic regression, two-hidden-layer feedforward neural networks, and autoencoder-pretrained models from scratch.
- Compared architectures across classification and regression datasets using repeated 5x2 cross-validation and task-appropriate performance metrics.
- Found that deeper networks generally improved classification performance and that autoencoder pretraining improved generalization on noisy nonlinear regression tasks.

Decision Trees with Reduced-Error Pruning

October 2025

Python | NumPy | pandas

- Implemented classification and regression trees using gain ratio and mean-squared-error split criteria.
- Added reduced-error pruning with held-out validation to control overfitting and improve model generalization.
- Reduced tree depth substantially with little or no loss in classification accuracy and improved error on noisy regression tasks.

Nonparametric Models: k-Nearest Neighbors

September 2025

Python | NumPy | pandas

- Built an end-to-end KNN pipeline supporting normalization, value-difference metrics for categorical features, Gaussian kernel regression, and null-model baselines.
- Implemented edited nearest-neighbor pruning to reduce training-set size while preserving predictive performance.
- Evaluated models with repeated 5x2 cross-validation across classification and regression datasets.

Personal Media Server

August 2025

Jellyfin | MakeMKV | Linux/Raspberry Pi | Bash

- Built a self-hosted Jellyfin streaming stack on a Raspberry Pi to consolidate and serve a personal media library.
- Created a Bash pipeline to automate disc handling, conversion to H.264, file movement, naming, and organization into a Jellyfin-compatible library structure.
- Used Linux command-line tools to implement logging, configure storage mounts and permissions, and clean empty work directories automatically.

Clue-Less Multiplayer Board Game

January-March 2025

Godot 4 | C# | Team of 5

- Designed and developed a simplified multiplayer interpretation of the classic board game Clue using Godot 4 and C#.
- Implemented gameplay systems and custom assets while collaborating within a five-person development team.
- Integrated networking libraries to support multiplayer interaction and coordinated system behavior.

2-D Platformer Prototype

November-December 2024

Godot 4 | GDScript

- Developed custom physics, collision detection, and a state-machine-driven gameplay loop in Godot 4.

TRIVIAGPT

June-August 2023

Java | libGDX | OpenAI API | MongoDB | Team of 6

- Architected a full-stack trivia game that generated dynamic questions from user-supplied categories through the OpenAI API.
- Built a RESTful service with a MongoDB backend for sensitive database operations, game-state data, and player information.
- Designed and implemented user-interface components, visual assets, and game logic in Java with libGDX.
- Coordinated project tasks, supported a GitHub feature-branch workflow and code reviews, and maintained project plans, requirements, design documentation, test plans, and the final report.

Custom Compiler

March-May 2023

C | Flex | Bison

- Designed a lexical analyzer and parser for a custom programming language, using regular expressions to tokenize source code and construct a parse tree.
- Implemented symbol, literal, and error tables to support syntax checking, semantic analysis, and detailed compiler diagnostics.
- Generated executable output and achieved greater than 95% pass rate on the automated test suite.

LEADERSHIP AND SERVICE

Eta Beta Chapter Member | Kappa Kappa Psi, National Honorary Band Fraternity

2009-2013

- Supported university-band concert preparation and coordinated community-service activities and event logistics.

Trumpeter | Marching Virginians

2009-2010

- Performed in more than 15 NCAA football halftime shows before crowds of approximately 60,000 while balancing rehearsals, performances, and coursework.